Staff Software Engineer, Safety

Location

Santa Clara

Employment Type

Full time

Location Type

On-site

Department

Software Engineering

OverviewApplication

Safety isn't a constraint — it's the foundation that enables robots to work seamlessly alongside people. At Cobot, we're redefining what safety means in human-robot collaboration by designing systems that are transparent, predictable, and deeply reliable in real-world environments.

As a Staff Safety Software Engineer, you'll help shape how safety is designed, tested, deployed, and continually improved across our robots. You will respond effectively to incidents, and maintain the rigorous standards required for certification and regulatory approvals.

This role offers the opportunity to influence not just the product, but the organizational culture around safety — partnering across teams, guiding technical decisions, and elevating our functional safety capabilities. If you're motivated by the challenge of enabling safe autonomy at industrial scale, we'd love to meet you.

Join us to reimagine the future of human-robot interaction.

Collaborative Robotics is a team of innovators and builders redefining the future of human-robot interaction. We are working to realize a world where robots are a trusted extension of your surroundings. They work, adapt, and react around you. Not the other way around.

This role is located onsite at our Santa Clara, CA headquarters. Cobot will offer a relocation stipend if you are relocating to join our Santa Clara office and currently live more than 50 miles outside of the office location.

Key Responsibilities:

- Architect and deliver safety-critical software and compute subsystems, including SIL-rated designs, documentation, and evidence for certification and regulatory review.
- Evaluate and integrate safety-rated hardware, software components, and development toolchains, ensuring performance, reliability, and compliance with functional safety standards.
- Analyze real-world and simulated operational data to assess safety performance, identify risks, and contribute to continuous fleet-level improvements and mitigations.
- Contribute to the product safety case and safety culture, partnering across engineering and operations to embed safety into design, development, and deployment.

Minimum Qualifications:

←

- 5+ years developing embedded or real-time software for safety-critical or regulated systems.
- Strong C/C++ skills for low-level and embedded development.
- Experience with RTOS, firmware bring-up, device drivers, and MCU peripherals.
- Proficiency in safety analysis methods (FMEA, FTA, HARA, STPA).

- Ability to produce high-quality documentation and follow structured, traceable development processes.
- Strong cross-functional communication skills.
- Comfortable working in a fast-paced startup and small, interdisciplinary teams.
- Must have and maintain US work authorization.

Preferred Qualifications:

- Advanced degree (Master's or PhD) in a relevant field.
- Hands-on work with IEC 61508 or comparable safety standards (ISO 26262, DO-178C, ISO 13849).
- Experience architecting SIL-rated compute or control subsystems and preparing certification evidence.
- Background evaluating safety-rated compute hardware, software components, and toolchains.
- Familiarity with safety case development, safety argumentation, and lifecycle safety artifacts.
- Experience with asymmetric multicore MCUs, redundancy techniques, or mixed-criticality systems.
- Knowledge of industrial robot safety standards (ISO 10218, ISO 12100, ISO 13849).
- Experience with regulatory engagement, audits, and safety documentation.
- Understanding of sensor fusion, perception, or robotics system integration.
- Experience promoting a safety-first culture through hazard analysis, incident response, and training.

The annual base salary range for this position is $205,000 - $235,000 plus equity and comprehensive benefits including medical/dental/vision plans, 401(k) plan, unlimited paid vacation time, paid sick leave, and parental leave. Our salary ranges are determined by role and experience level. The range reflects the minimum and maximum target for new hire salaries for the position in the noted geographic area. Within the range, individual pay is determined by additional factors, including job-related skills, experience, and relevant education or training.

Cobot is an Equal Opportunity Employer. All qualified applicants will receive consideration for employment without regard to legally protected characteristics. We are committed to providing reasonable accommodations for candidates with disabilities in our recruiting process. If you need any assistance or accommodations due to a disability, please let your recruiter know.

To all recruitment agencies: Collaborative Robotics does not accept agency resumes. Please do not forward resumes to our employees. Collaborative Robotics is not responsible for any fees related to unsolicited resumes.

Apply for this Job